

Title page

Manuscript title

**Effects of resisted sprint training on acceleration with three different loads
accounting for 5%, 12.5% and 20% of body mass.**

Brief running head

Effects of resisted sprint training with three different loads

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Abstract

The optimal resisted load for sprint training has not been established yet, although it has been suggested that a resistance reducing the athlete's velocity by more than 10% from unloaded sprinting would entail substantial changes in the athlete's sprinting mechanics. This investigation has evaluated the effects of a 7-week, 14-session sled resisted sprint training on acceleration with three different loads according to a % of body mass (BM): low load (LL: 5% BM, n = 7), medium load (ML: 12.5% BM, n = 6) and high load (HL: 20% BM, n = 6), in young male students. Besides, the effects on untrained exercises: vertical jump (CMJ), loaded vertical jump (JS) and full squat (SQ) were analyzed. The three groups followed the same training program consisting in maximal effort sprint accelerations with the respective loads assigned. Significant differences between groups only occurred between LL and ML in CMJ ($p < 0.05$), favoring ML. Paired t-tests demonstrated statistical improvements in 0-40 m sprint times for the three groups ($p < 0.05$), and in 0-20 m ($p < 0.05$) and 0-30 m ($p < 0.01$) sprint times for HL. Sprint times in 10-40 m ($p < 0.01$) and 20-40 m ($p < 0.05$) were improved in LL. Time intervals in 20-30 m and 20-40 m ($p < 0.05$) were statistically reduced in ML. As regards the untrained exercises, CMJ and SQ for ML and HL ($p < 0.05$) and JS for HL were improved. The results show that, depending on the magnitude of load used, the related effects will be attained in different phases of the 40 m. It would seem that to improve the initial phase of acceleration up to 30 m, loads around 20% of BM should be used, whereas to improve high-speed acceleration phases, loads around 5 to 12.5% of BM should be preferred. Moreover, sprint resisted training with ML and HL would enhance vertical jump and leg strength in moderately trained subjects.

Keywords

Pull sled - velocity loss - specific strength training - running.

INTRODUCTION

The ability to run at maximum speed for short periods of time and the ability to jump are important performance factors, both in individual sports such as athletics, and in numerous team sports such as football, basketball, and rugby. Most sprint training programs include strength-specific exercises, in which the athlete reproduces the technical movement with an added resistance (resisted sprint training) (16).

The main objective of resisted training exercises is to produce an additional load in the sprint muscles involved, as it is considered that this additional load produces a neural activation and a larger recruitment of fast motor units (7, 9, 10). Most studies have investigated the influence of resisted sprint training on physical education students (19, 31) and field sports athletes such as lacrosse players (6), football (or soccer) players (28) and rugby players (15). The training frequencies in those studies were 2–3 days per week over 6–8 weeks; the weighed sled resistances were 7–13% of the body mass. The sprint training and testing distances ranged from 9.1 to 55-m, and usually the number of training sets varied from 3 to 9 per session (16). Some studies have found that resisted sprint training increases sprint speed; although there are no significant differences for sprint speed when compared with normal sprint training (5, 19, 23, 28). However, Harrison and Bourke (15), and Zafeiridis et al. (31) showed that resisted training increased sprint speed more than normal sprint training. These improvements were mostly visible during the initial acceleration phase (31); nonetheless, Spinks et al. (28) didn't find any statistical improvements in the above-mentioned phase. Resisted sprint training has not been reported to lead to better results for distances >20 m. In fact, normal sprint training has been shown to produce better results than those of resisted sprint training for distances of ≥ 20 m (19, 31).

Research on resisted sprint training methods has mainly dealt with towing weighted sleds (6, 15, 28, 31), weighted vests (6) and parachutes (24). Most of this research compares the effect of different types of resisted speed training with normal running training (28, 30, 31), or with assisted sprint training (19). We have found only one longitudinal study comparing two different loads [heavy (30% decrease in sprint velocity) vs. light load (10% decrease in sprint velocity)] (18). Besides, we have found just one study (28) analyzing the effects of a weighted sled-training period on CMJ. In that study, statistical improvements were observed in CMJ after 8 weeks of resisted sprint training with a 10% load of body mass.

It has been suggested that to achieve a positive effect, without changing the athlete's sprinting mechanics, the resistance when towing a sled should not reduce the athlete's velocity by more than 10% from unloaded sprinting (1, 17, 20, 21). Nevertheless, the optimal load for resisted sprint training has not been established yet by longitudinal studies.

Therefore, the objective in the present study was to compare the effects of resisted sprint training on acceleration with three different loads accounting for 5%, 12.5% and 20% of the body mass. Besides, a secondary question addressed in this study was to analyze the effects of resisted sprint training on untrained exercises like CMJ, JS, and SQ.

METHODS

Experimental Approach to the Problem

A longitudinal experimental study comparing three different loads of resisted sprint training on acceleration and strength variables was carried out. The participants were randomly assigned to one of three intervention groups: LL (5% of BM), ML (12.5% of BM) and HL (20% of BM). Loads equivalent to 5%, 12.5% and 20% of BM entailed respectively a mean velocity loss of 5.6%, 10% and 15.5%. The participants performed 0-40 m sprint test, with running times being measured each 10 m, and strength tests (CMJ, JS and SQ), before and after a 7-week resisted training period of 2 non-consecutive sessions/week, i.e. a total of 14 sessions. The three groups completed identical training activities (same number of sets, repetitions and running distances). The testing was performed with two pre-training and two post-training sessions. Session one consisted of a 0-40 m sprint test and session two consisted of CMJ, JS and SQ tests. Pre and post-tests were performed under the same conditions: time of the day, facilities, and moment of the week. Sprint tests were conducted on a synthetic running track in an indoor hall. Differences between groups were determined, and intra-group differences were compared between pre-test and post-test.

Subjects

Nineteen male students in sport science, physically active, were selected for this study. The participants were divided into three training groups: low load group (LL, n=7, age: 21.9 ± 2.27 years, height: 180.9 ± 6.78 cm, weight: 75.8 ± 10.73 kg), medium load group (ML, n=6, age: 20.8 ± 2.04 years, height: 173.8 ± 4.58 cm, weight: 66.8 ± 8.53

kg) and high load group (HL, n=6, age: 19.8 ± 1.60 years, height: 175.4 ± 6.79 cm, weight: 70.2 ± 11.91 kg). None of the participants had previous experience with specific sprint training. To take part in the study, the participants had to fulfill the following requirements: 1) to be healthy and not to suffer any illness that could be a risk under intense physical activity; 2) to be generally well-trained; and 3) not to perform any physical training activity other than that of the resisted sprint training program. All the participants were fully informed about procedures and their informed consent was obtained prior to the first tests. The experiments were carried out in accordance with the ethical standards laid down in the 1964 Declaration of Helsinki.

Procedures

Test preparation. During the first testing day (0-40 m sprint test), all the participants completed a 15 minutes standard warm-up consisting of 8 minutes of low-intensity jogging, stretching, and one 0-40 m sprint at 80% effort and two 0-20 m at 90% effort. The second day (strength tests), in addition to the standard warm-up, the participants did 5 CMJ jumps and SQ without load.

Squat and jump measurements. The protocols about the test performance were fully explained to all participants before starting the test; and these completed two sessions to become familiarized with the exercises. The CMJ and JS were performed on an Optojump (Microgate, Italy) Measurement System. This system has proved to produce reliable results for jumping activities (11). The CMJ was performed with both hands on the waist. The participants were required to do 3 trials, mean height being scored. The JS test was carried out with progressive loads from the bar weight (20 kg) up to the load allowing the subject to jump up no more than 20 cm high or less. For data analysis,

mean heights of pre-test and post-test common loads were adopted. As for the SQ test, the movement velocity of each load in the concentric phase was measured. A dynamic measurement system (T-Force System; Ergotech, Murcia, Spain) automatically calculated the relevant kinematic parameters of every repetition. This system consists of a linear velocity transducer. Instantaneous velocity was sampled at a frequency of 1000 Hz. Both the reliability and validity of the system have already been established (26). In the SQ test, the participants started from the upright position with their knees and hips fully extended in a Smith machine (Multipower Fitness Line, Peroga, Spain) that ensures a smooth vertical displacement of the bar along a fixed pathway, the barbell resting across their back at the level of the acromion. The participants were always required to execute the concentric phase of SQ in an explosive manner, at maximal intended velocity. After a warm-up, initial load was set at 20 kg for all participants and was gradually increased in 10 or 5 kg increments until the attained mean propulsive velocity (MPV) was $<0.8 \text{ m}\cdot\text{s}^{-1}$. This velocity of $0.8 \text{ m}\cdot\text{s}^{-1}$ corresponds approximately to 70% RM and was considered sufficient load to analyze the effects of the training program on leg strength (25). For each load, the repetition correctly performed at the highest velocity was scored. The velocities reported in the present study correspond to the mean velocity of the propulsive phase (27) for each common load, from 20 kg to the load of $0.8 \text{ m}\cdot\text{s}^{-1}$.

Sprint testing. Two 0-40 m trials were performed. To time the sprint trials, 5 couples of photocells (Polifemo Radio Light, Microgate, Italy) based on a radio impulse transmission system and a reflection system were employed. This device is completely accepted and widely used in electronic timekeeping systems (8). Runs were performed from static biped start position with the start line located 1 m behind the start photocell. The rest of the photocell pairs were placed at 10 m, 20 m, 30 m and 40 m. The best time of the two 0-40

m trials was scored. The rest period between sets typically lasted about 6 minutes, which is sufficient for full recovery from repeated maximal sprints of short duration (14).

Training protocol. The participants completed two training sessions per week for a period of 7 weeks, completing 14 training sessions. The training sessions were held with an interval of at least two days. The participants began with the same standardized warm-up before the training session, which included 8 minutes of low-intensity jogging, stretching, and one 0-40 m sprint at 80% effort and two 0-20 m sprints at 90% effort. Each subject trained with the load previously determined according to the load group he had been assigned to. Training sessions consisted of 4-8 sets ranging from 20 to 35 m, always at maximal effort, with 3-5 minutes recovery time (Table 1).

(Table 1 about here)

Statistical Analyses

The variables are reported as mean $\pm$ standard deviation (SD). The reliability of all measures was set with the intraclass correlation coefficient (ICC) and the coefficient of variation (CV). Confidence interval (CI) was set at 95%. The correlations between variable changes were analyzed using the Pearson correlation coefficient. The differences between groups were determined through the analysis of variance with test 1 as co-variable (ANCOVA). Significant differences were determined by a Scheffé post-hoc test. A Student's dependent t-test for paired samples (CI: 95%) was performed to compare the pre-test with post-test intra-group differences. Statistical significance was set at $p \leq 0.05$. SPSS for Mac (release 20.0.0) was used for all statistical analyses.

RESULTS

Reliability was set with ICC and CV for all the distances and jumps. In the study, the sprints and jumps were generally reliable. For all running distances ICC was higher than 0.9 except in 0-10 m (ICC: 0.87). The maximum CV value for running distances was 2.61 (in 20-30 m), and the minimum CV value was 0.58 (in 0-40 m). For vertical jump (CMJ), the ICC value was 0.99, and the CV value was 1.74. For the loaded vertical jump (JS), the ICC and CV values were 0.97 and 3.98 for JS_{20kg}, and 0.92 and 5.54 for JS_{30kg}, respectively.

All groups reduced their times in the 0-10 m, 0-20 m, 0-30 m and 0-40 m sprints but the changes were only statistically significant in the 0-40 m sprint ($p < 0.05$) for the 3 groups and in the 0-20 m ($p < 0.05$) and 0-30 m sprints ($p < 0.01$) in the HL group (Figures 1A, 1B, 1C). All groups reduced their times in 10-40 m, 20-30 m, 20-40 m and 30-40 m intervals. These changes were significant in 10-40 m ($p < 0.01$) and 20-40 m ($p < 0.05$) intervals in LL (Figures 1D, 1F). In ML the significant improvements were in 20-30 m and 20-40 m ($p < 0.05$) sprint interval times (Figures 1E, 1F), but there were no significant differences in HL in any of these interval distances.

With respect to the CMJ, statistically significant improvements were only seen in ML and HL ($p < 0.05$) (Figure 2A). In JS, although all groups increased the height of the jump, significant statistical differences were only observed in HL ($p < 0.05$) (Figure 2B). Significant differences between groups only occurred between LL and ML in CMJ ($p < 0.05$), favoring the latter (Figure 2A). For the SQ exercise, the 3 groups increased

the average MPV of the common loads but only significantly in ML and HL ($p < 0.05$) (Figure 3).

The correlations between the changes in times in 0-20 m, and those in 10-20 m ($r = 0.46$, $p < 0.05$), 10-30 m ($r = 0.25$, not significant: ns), 10-40 m ($r = 0.12$, ns), 20-30 m ($r = 0.004$, ns), 20-40 m ($r = -0.19$, ns), and 30-40 m ($r = 0.07$, ns) showed a downward trend as the distance covered increased.

DISCUSSION

The results of our study show that, after a 7-week period of resistance sprint training with sleds (2 sess. /week), all groups improved statistically the time in 0-40 m sprint. The HL group also improved statistically the times in 0-20 m and 0-40 m, but not in 0-10 m. The ML and LL groups improved statistically the interval times (LL: 10-40 m and 20-40 m; ML: 20-30 m and 20-40 m), but not in 0-10 m, 0-20 m, and 0-30 m.

Our results partially coincide with those of Zafeiridis et al. (31) who, after an 8-week of resistance sprint training program (3 sessions/week), also found significant improvements in 0-10 m and 0-20 m sprint times in the group that trained with sleds. Since all of the participants, there, trained with the same absolute load (5 kg), it is reasonable to think that the relative resistance to movement was different in each subject. By contrast, in our study, three different loads were used and adjusted according to body weight. There is another difference in the results because, in our case, it was the HL group (20% BM, with an average of 14 kg), which improved significantly in 0-20 m and 0-30 m sprint times, that is, when acceleration is measured from zero velocity. Likewise, the LL (5% BM, average 3.8 kg) and ML groups (12.5% BM,

average 8.4 kg), who trained with loads closer to those of the above-mentioned study (31) improved sprint times significantly in a flying phase, when velocity is higher (LL in 10-40 m and 20-40 m, and ML in 20-30 m and 20-40 m). These discrepancies in the range in which significant gains were seen could be due to the total number of meters covered during training in both studies. The distance covered was much higher in the study by Zafeiridis et al. (31) (280 m per session, three times a week, of which 71% were a 50 m series), while in our case there were two sessions per week and the distance covered ranged between 100 and 210 m, the latter being done on only one occasion. The explanation for this difference could be the greater distance covered in training, which may have caused excessive fatigue. As it has been observed, the duration of muscular contraction increases with fatigue (2, 12, 13), which might have reduced the rate of force development (RFD) and produced a greater contact time in the sprint (4, 29). Therefore, the effect of fatigue is expressed especially in the maximum speed phase, in which the contact time is reduced and the RFD is more critical (29). This could be the reason why significant improvements were found in the maximum speed phase with low-load training in our case but not in the afore-mentioned study.

Spinks et al. (28) found significant improvements in 0-15 m sprint time after eight weeks of resisted sprint training (2 sessions/week) with a load (10% of BM) similar to that used by our ML group (12.5% BM), but the improvements in the results were seen in the initial phase of the sprint, which did not occur in our study. It should be underlined that in the study by Spinks et al. (28) the participants did additional exercises, such as football training, strength training, plus a match per week, then the observed discrepancy in the results could be attributable to many different variables; therefore, both studies are not directly comparable. Likewise, the effect of the resisted

sprint training on the flying phases cannot be compared between the two studies since Spinks et al. (28) only measured the time in 0-15 m.

Kawamori et al. (18) also found significant improvements in 0-10 m sprint time after eight weeks of resisted sprint training (2 sessions/week) in both light (10% decrease in sprint velocity) and heavy loads group (30% decrease in sprint velocity). The improvements were higher in the heavy group, and significant improvements were found in 0-5 m sprint time only for this group, but this heavy group cannot be compared with our HL group since the training load was higher than the one used by our HL group (15.5% decrease in sprint velocity). The light group used a load with 10% decrease in sprint velocity which is similar to that used by our ML group; but although our ML group reduced time in all distances, those improvements were not significant in 0-10 m. It should be pointed out that the participants in that study performed not only weighted sled towing but also a considerable volume of unresisted sprints both within the training program as part of the warm-up and sled mass adjustment procedure, and outside the program during skill practice sessions and competitions.

Another recent study (30) compared the effects of combined sled towing and sprint training with traditional sprint training groups on 0-10 m and 0-30 m sprints in professional rugby players during a 6-week training period (2 sessions/week). In that study, the 0-10 m and 0-30 m sprint times were significantly improved in the combined sled towing and sprint training group. The sled load was 12.6% BM, similar to our ML group. Those results support ours; however, they do not indicate to what extent the improvements were due to the resisted sprint training or to normal sprint training. Instead, our study shows that the improvements were exclusively due to the resisted sprint training with load. In this sense, our study differs from West et al.'s (30) since it

provides specific information about the effect of the exclusive use of resisted sprint training with load.

The results of Harrison and Bourke (15) differ from ours as, after 6 weeks of resisted sprint training (2 sessions/week) with a load of approximately 13% BM, similar to that used by our ML group, no significant gains were found in 0-30 m sprint times; still, these appeared in 0-5 m and 0-10 m sprint times. Although the distances for which better results were observed correspond approximately to those at which our HL group also improved; these results are not directly comparable since the participants, who were rugby players, performed speed-training drills every week as part of their habitual training.

The results obtained by Kristensen et al. (19) are similar to those obtained by the LL and ML groups in our study since, in both cases, no significant improvements were found in 0-20 m acceleration times. The similarity between the results could be due to the fact that the resistance to movement was similar, as the velocity loss during the training sprints was almost the same in both studies. The training load used by Kristensen et al. (19), consisting of a traction system in the direction of movement without towing, entailed a velocity loss of 8.5% with respect to the speed without load. This velocity loss would correspond, in our study groups, to a medium load between LL (5.6% velocity loss) and ML (10% velocity loss).

Our results partially coincide with those of Clark et al. (6), in which the group that trained with towing ($\approx 10\%$ BM) over 7 weeks obtained trivial improvements in times in 18-55 m sprint interval. These results are very similar to those of our study for the ML

group (12.5% BM), which improved significantly ($p < 0.05$) in 20-40 m sprint interval time. However, it is not possible to compare the acceleration phase, which was not measured in that study (6).

According to the sprint results, the improvement of the HL group in 0-40 m sprint time was due to enhanced 0-20 m and 0-30 m acceleration phases (Figures 1A, 1B), since, for this group, the sprint times in the 20-30 m, 20-40 m and 30-40 m flying intervals were stable with respect to the initial test. These participants, having trained with greater loads, i.e. at lower speeds during training, didn't improve times in those flying intervals with respect to their initial results. These significant improvements in the HL group in 0-30 m ($p < 0.01$) were reinforced by the fact that all participants (6 of 6, 100%) achieved a better time in 0-30 m (Figure 1B). By contrast, the performance in 0-40 m sprint times in LL and ML groups was essentially due to improvements in the 10-40 m and 20-40 m flying phases for LL, and 20-30 m and 20-40 m for ML, and not in the acceleration phase 0-30 m. This is coherent with the above, since these participants trained with a smaller load (5% and 12.5% of BM), which resulted in their losing less speed in each training series. These loads were perhaps too light to allow improvements in the acceleration time in 0-30 m, in which the force application time is greater.

It has been observed that during the acceleration phase (from a standing start), the participants who reduce most the time in 0-20 m tend to be the same as those who reduce least the time in the flying phases, but these changes are not significant. This would suggest that the improvement achieved in the first 0-20 m is unrelated to the improvements achieved in the flying phases (10-40 m, 20-30 m, and 20-40 m), where speed was higher.

As regards the jump and strength variables (CMJ, JS, and SQ), for which no specific training was done, all groups improved, although no significant differences between pre and post tests were found in LL. ML and HL groups improved significantly in CMJ and SQ; besides, the HL group also improved in JS. Sled resisted training with 12.5% and 20% of BM turned out to be true strength training in moderately trained subjects. These results suggest the following positive lineal tendency: the higher the training load in the sled (between 5% and 20% of BM), the bigger the effect on these untrained exercises. This could be explained by the fact that the “explosive force” of the knee extensor muscles has traditionally been closely related to jump and acceleration in the sprint (2, 3, 22).

As a conclusion, these results highlight that the different acceleration phases can be improved without parallel use of training without loads. So, it would appear that training with sleds for 7 weeks with the three loads used, and without additional unloaded sprint training would improve the 0-40 m sprint performance (LL, ML, and HL). Such improvement could be related to the use of different training loads. As a matter of fact, the LL and ML groups reduced times significantly in the flying phases (LL: 10-40 m and 20-40 m, ML: 20-30 m and 20-40 m), whilst the HL group reduced times in the 0-20 m and 0-30 m accelerations, but not in the flying phases. Furthermore, resisted sprint training tends to produce improvements in exercises that were not trained such as CMJ, JS and SQ with high loads (20% BM) and in CMJ and SQ with medium loads (12.5% BM). The higher the load within the range of loads used in our study, the greater the improvement.

PRACTICAL APPLICATIONS

The results of our study would suggest that in order to improve the acceleration phase over short distances it is advisable to train with different towed loads. To improve the initial acceleration phase up to 30 m, it would be advisable to train with loads about 20% of BM. However, to improve the flying phases within 40 m, loads between 5% and 12.5% should be used.

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FIGURE LEGENDS

Figure 1. Mean sprint times and individual values for each subject in the unloaded running sprinting tests: (A) 20 m; (B) 30 m; (C) 40 m; (D) 10-40 m split; (E) 20-30 m split; (F) 20-40 m split. Training groups: LOW load (LL), MEDIUM load (ML), HIGH load (HL).

Significant differences from PRE- to POST-training: * $p < 0.05$; ** $p < 0.01$

Figure 2. Mean jump height and individual values for each subject in: (A) CMJ; (B) JS.

Significant difference from PRE- to POST-training: * $p < 0.05$.

Significant differences between-groups

Figure 3. Mean propulsive velocity (MPV) attained with common absolute loads at the PRE- and POST-tests in the full SQ exercise, and individual values for each subject.

Significant difference from PRE- to POST-training: * $p < 0.05$

TABLE 1. Resisted sprint training program for the three resisted sprint groups: LL (n=7), ML (n=6), HL (n=6).

Week	Session	Distance (m)	Repetitions	Total distance (m)	Total Week Distance (m)	Recovery (min)
1	1	20	5	100	250	3
	2	30	5	150		4
2	3	20	6	120	300	3
	4	30	6	180		4
3	5	20	8	160	340	3
	6	30	6	180		4
4	7	20	8	160	335	3
	8	35	5	175		5
5	9	20	8	160	370	3
	10	35	6	210		5
6	11	20	8	160	280	3
	12	20	6	120		3
7	13	30	4	120	240	4
	14	20	6	120		3





